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Neogene Tectonic Rotation of the San Gabriel Region, California, Suggested by Paleomagnetic Vectors

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Previous paleomagnetic work in the western Transverse Ranges and the islands offshore of California has suggested that these regions have experienced large clockwise rotations since middle Miocene time. This study is aimed at testing how far east these rotations extend into the continent. Neogene volcanic rocks from the region bounded by the San Gabriel and San Andreas faults were selected for paleomagnetic study. The sampled units are lavas from the late Oligocene to early Miocene continental Vasquez Formation, and water-laid tuff beds from the middle to late Miocene Mint Canyon Formation. These rocks possess a stable remanent magnetization which was acquired soon after deposition and is believed to be primary. Seventy-four lava flows from eleven locations in the Vasquez Formation were studied. The average direction for seven locations near the center of the study area suggests that this area has rotated a net clockwise amount of 37.1° ± 12.2° since the early Miocene. There is no suggestion of north or south translation. A location west of the central region is rotated almost 90° clockwise, although this is believed to be a local effect. Detailed magnetic stratigraphy at the western and central location can be correlated over 6 km. Locations in the east near the San Andreas fault yielded highly dispersed directions which cannot now be interpreted in terms of rotation. The mean paleomagnetic direction from six sites (four different beds) in the Mint Canyon formation west of the Vasquez study area is deflected in a counterclockwise sense by 16° ± 30°. These data are consistent with a proposed model in which the San Gabriel region first was rotated clockwise 53° (37° + 16°) in the early Miocene and later rotated 16° counterclockwise as it slid northward into the leftward bend region of the San Andreas fault. The clockwise rotation is believed to be due to dextral shear within the Pacific-North American plate shear zone; counterclockwise rotation in the Mojave region is believed to be responsible for the bend in the San Andreas fault.

INTRODUCTION

The Transverse Ranges of southern California have a west trending structural and stratigraphic grain which cuts across the northwest grain of California (Figure 1). Anomalous west trending features include high-angle left-lateral and reverse faults (i.e., Big Pine, Santa Ynez, Oak Ridge, and Malibu Coast faults) [*Crowell*, 1968; *Jahns*, 1973; *Jennings*, 1977], Mesozoic metamorphic basement trends [*Jones et al.*, 1976], the Mesozoic and Paleogene forearc basin [*Jones et al.*, 1976; *Howell*, 1974; *Crouch*, 1979], and mid-Tertiary continental basins [*McCracken*, 1972; *Bohannon*, 1975]. In addition, detailed seismic studies suggest that much of the Transverse Ranges are "rootless" and are the site of a west trending upper mantle ridge [*Hadley and Kanamori*, 1977].

Many of the west trending features of the Transverse Ranges are thought to have been tectonically rotated clockwise in Neogene time from an originally more northerly orientation; these areas are shown shaded in Figure 1 [*Kamerling and Luyendyk*, 1979; *Luyendyk et al.*, 1980]. This idea is suggested from paleomagnetic studies of mainly Miocene volcanic rocks of the Santa Monica Mountains and the northern Channel Islands [*Kamerling and Luyendyk*, 1979, this issue; *Kamerling*, 1985] and of the Miocene Monterey Formation [*Hornafius*, 1984, this issue] which consistently yield clockwise-deflected magnetic declinations of up to 100°. These studies have provided valuable insight into the tectonics of the western Transverse Ranges. In this paper, paleomagnetic data are presented from the mid-Tertiary Vasquez and Miocene Mint Canyon formations of the central Transverse Ranges in an effort to characterize better the eastward extent, timing, and magnitude of these major clockwise rotations.

GEOLOGIC SETTING

The Vasquez and Mint Canyon formations are found in the San Gabriel block (the lens-shaped region bounded by the San Gabriel and San Andreas faults; Figure 1). Basement within the San Gabriel block comprises a diverse suite of Precambrian gneiss, granulite gneiss, an anorthosite-syenite complex, Mesozoic granodiorite and granite, and Mesozoic (?) schist [*Ehlig*, 1975; *Silver*, 1971]. A transverse structural grain is imparted to this region by a series of east-northeast trending left-slip faults (Figure 2). The Vasquez Formation was deposited in three structural basins (Charlie Canyon, Texas Canyon, and Soledad basins) which presently are oriented roughly ENE.

Vasquez Formation

The Vasquez Formation is a late Oligocene to early Miocene continental red bed and volcanic sequence [*Oakeshott*, 1958; *Muehlberger*, 1958; *Crowell*, 1973; *Bohannon*, 1975]. It comprises up to 1500 m of alluvial fan, fluvial, and playa deposits. Near the base of the formation it is up to 1300 m of basalt and andesite flows and minor agglomerate [*Oakeshott*, 1958; *Bohannon*, 1975]. Three

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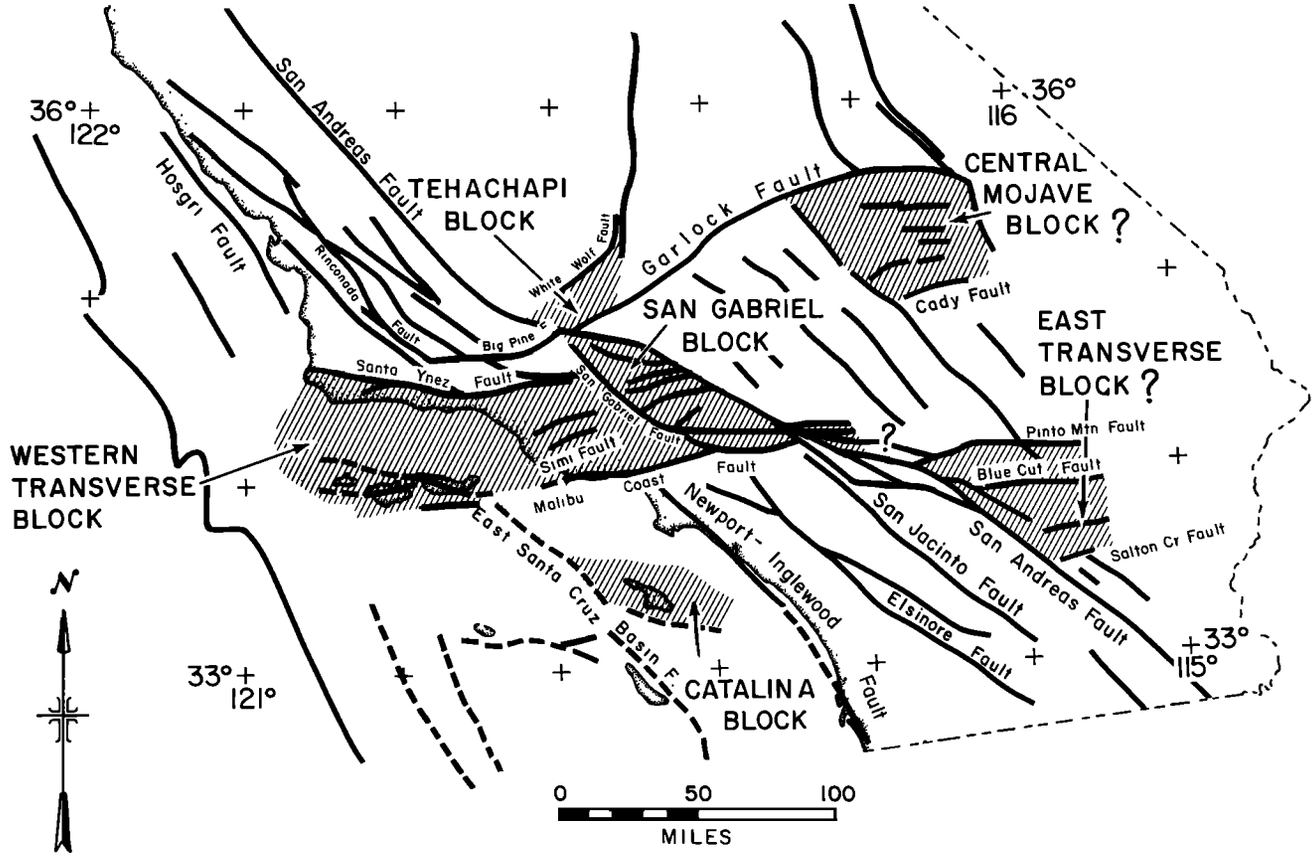


Fig. 1. Fault map of southern California [after *Jennings*, 1973]. The shaded areas are regions where Neogene rotations are suspected from paleomagnetic data or geologic structure (question marks). Modified from *Luyendyk et al*, [1980, Figure 2].

K-Ar dates from these basalt flows yield early Miocene ages of 23.9 ± 0.8, 24.9 ± 2.1, and 20.9 ± 0.8 Ma [*Crowell*, 1973; *Crowe et al.*, 1979]. The Vasquez Formation unconformably overlies the crystalline basement complex of the San Gabriel block, and is in turn angularly overlain by the lower to upper Miocene Tick Canyon and Mint Canyon formations [*Oakeshott*, 1958].

Mint Canyon Formation

The Mint Canyon Formation ranges in age from middle to late Miocene and comprises up to 1800 m of fluvial and lacustrine deposits [*Jahns*, 1939, 1940; *Oakeshott*, 1958; *Ehlert*, 1982]. It conformably overlies the early to middle Miocene Tick Canyon Formation and has an upper contact which is in part conformable and in part angularly uncon-

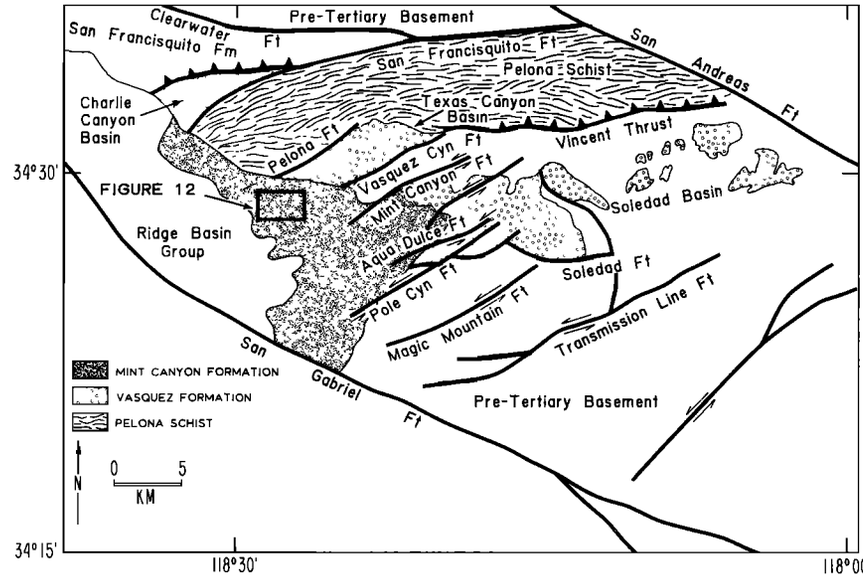


Fig. 2. Generalized geologic map of part of the San Gabriel block [from *Jennings and Strand, 1969; Jennings, 1977*].

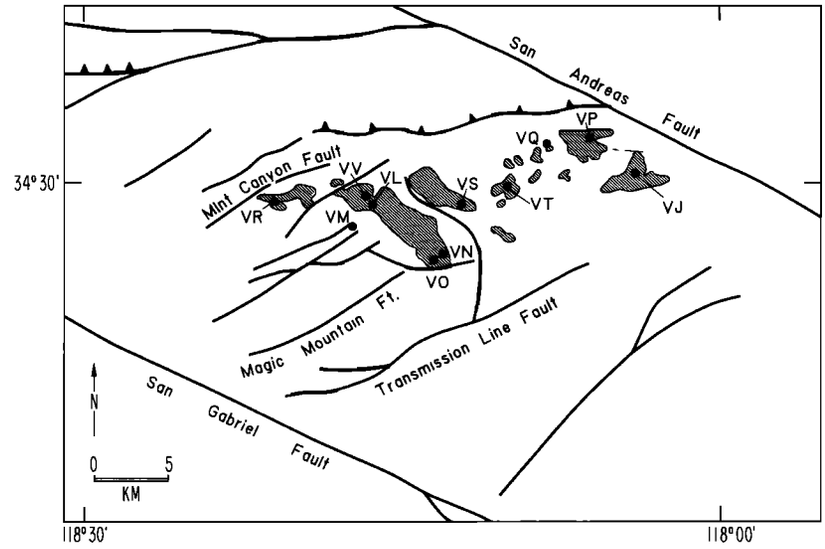


Fig. 3. Paleomagnetic sampling locations in the Vasquez Formation. The volcanic member is shaded.

formable with the overlying upper Miocene marine Castaic Formation [*Ehlert*, 1982]. The Mint Canyon Formation has yielded two distinct vertebrate fauna. It is separated by *Durham et al.* [1954] into lower (Barstovian) and upper (Clarendonian) members. Two zircon fission track dates from tuff beds of the upper member are 10.1 ± 0.8 Ma and 11.6 ± 1.2 Ma (J. D. Obradovich and T. H. McCulloch, personal communication, 1982).

The rocks of the lower Mint Canyon Formation are predominantly conglomerate and sandstone deposited in alluvial-fan and alluvial-wash environments [*Oakeshott*, 1958; *Ehlert*, 1982]. The rocks of the upper member include some conglomerates but are notably finer-grained than the lower member. These fine-grained rocks are predominantly white and gray tuff and thin-bedded mudstone and sandstone that were deposited in a shallow lake [*Oakeshott*, 1958; *Ehlert*, 1982]. The paleomagnetic study was conducted on the tuffs of the upper member.

PALEOMAGNETISM

A portable gas-powered rock drill was used to obtain 2.5-cm-diameter core samples. Each core typically yielded two 2.5-cm-long specimens. From 3 to 15 (typically six) samples were collected per unit. Magnetic orientations were taken with a Brunton compass, and periodically, sun sights were used to check for local magnetic aberrations. For structural corrections, attitudes generally were taken on bedding surfaces of fine-grained sedimentary rocks, but within the volcanic member of the Vasquez Formation it was sometimes necessary to sight along flow contacts.

Three magnetometers were used to measure the samples: a Schonstedt SSM-1A spinner magnetometer at the University of California, Santa Barbara (UCSB), and two Superconducting Technology cryogenic magnetometers, one at UCSB and another at Stanford University. Standards were run on all magnetometers, and no significant differences in either magnetic directions or intensities were observed.

The samples of the Vasquez Formation and white tuffs of the Mint Canyon Formation were subjected to stepwise alternating field (af) demagnetization. One or two pilot samples from each unit were demagnetized in 5 mT increments to 40 mT and then in 10 mT increments to 80 mT. Where warranted by stable behavior, abbreviated demagnetization sequences were used to clean the remaining samples of each unit. The gray tuffs and representative samples of the white tuffs of the Mint Canyon Formation were cleaned using stepwise thermal demagnetization. A Schonstedt TSD-1 tube-style oven was used. Each sample was demagnetized at 50°C intervals from 150°C or 250°C to 550°C, with continued demagnetization to 580°C or 610°C.

Stable directions were selected with the aid of orthogonal vector component plots [*Zijderveld*, 1967] and stereoplots. Unit means, means of several units, confidence limits (α_{95}), and precision parameters (κ) were calculated using the methods of *Fisher* [1953]. For any given mean paleomagnetic direction the uncertainty in inclination (ΔI) is ± α_{95} and in declination (ΔD) is ± $\sin^{-1}$ ($\sin \alpha_{95}$ / cos I) (where I is the inclination). The difference in inclination (i) and declination (d) and the associated uncertainties (Δi, Δd) between any two mean paleomagnetic directions (I_1, D_1, ΔI_1, ΔD_1) and (I_2, D_2, ΔI_2, ΔD_2) are calculated as follows:

$$d = D_2 - D_1 \tag{1}$$

$$i = I_2 - I_1 \tag{2}$$

$$\Delta d = (\Delta D_1^2 + \Delta D_2^2)^{1/2} \tag{3}$$

$$\Delta i = (\Delta I_1^2 + \Delta I_2^2)^{1/2} \tag{4}$$

[after *Wilson and Cox*, 1980]. For comparison of the mean paleomagnetic vectors with the appropriate reference pole, the method of *Beck* [1980] was used to determine the uncertainties in rotation, ΔR and flattening, ΔF. However, *Demarest* [1983] has shown that these limits are usually too large and should be contracted by about 20%. The method of *Watson* [1956] was used to test two sample populations for uniqueness. The reference poles used are those of *Irving* [1979].

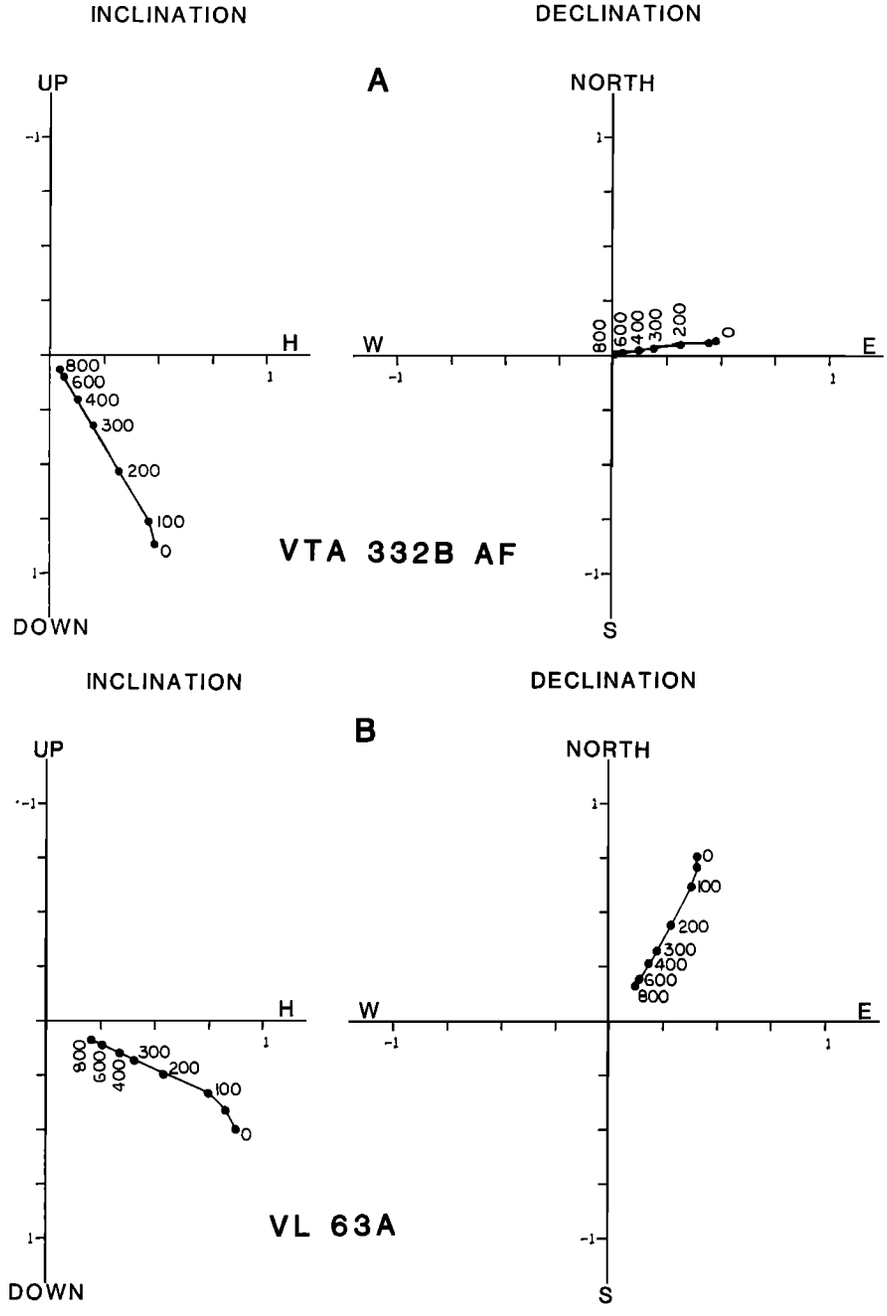


Fig. 4. Orthogonal vector component plots illustrating af demagnetization of Vasquez volcanics. Numbers are the demagnetizing level in milliTeslas. The directions of most samples are stable in demagnetization fields greater than 20 mT. Sample shown in Figure 4*a* is from location VT, and sample shown in Figure 4*b* is from location VL.

PALEOMAGNETISM OF THE VASQUEZ FORMATION

At eleven locations in the volcanic member of the Vasquez Formation, 415 oriented samples from 74 volcanic flows were collected (Figure 3). The units at locations VL, VN, VO, VR, VS, VT, and VV all were sampled at the same general stratigraphic interval (see map of *Oakeshott* [1958]), and some flows may be sampled more than once. The rocks sampled at location VM are approximately 300 m stratigraphically above these other flows (the intervening rocks are sandstones and conglomerates). The volcanic rocks of locations VJ, VP, and VQ rest directly on basement and cannot be observed interfingering with the flows sampled to the west. All of these volcanic units are thought to be of roughly the same age and genetically related [*Oakeshott*, 1958; *Bohannon*, 1976].

The Vasquez samples were subjected to alternating field demagnetization and typical behavior is shown in Figure 4. Most samples had only a minor present-day viscous remanent magnetization (VRM) overprint which was removed easily by a demagnetizing field of 10 to 20 mT. Demagnetization in fields between 20 and 100 mT generally yielded very little directional change and this stable direction is interpreted as the original thermal remanence

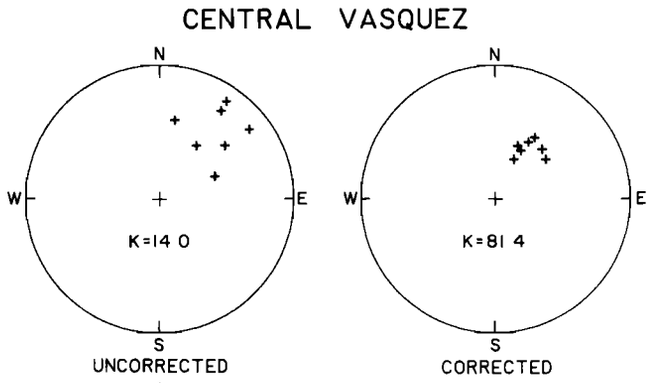


Fig. 5. Equal-area stereoplots illustrating positive fold test for the mean directions of seven locatins (VL, VM, VN, VO, VS, VT, and VV; Figure 3). All points are lower hemisphere. Before structural correction, $\kappa = 14.0$, and after correction, $\kappa = 81.4$.

(TRM). The median destructive field typically was between 30 and 50 mT. The hysteresis parameters and isothermal remanent magnetization (IRM) acquisition of samples from 20 units indicate that hematite is not present in significant amounts (samples saturated in fields of 250 mT), the magnetic particles are dominantly of pseudo-single domain size, and the magnetic properties are relatively uniform throughout the stratigraphic section and the sampling area [*Terres*, 1984].

The magnetization of the volcanic rocks was acquired prior to folding. The clustering of the mean directions of seven locations in the central Vasquez area is improved after a correction for local tilting and folding is applied (Figure 5). The precision parameter κ increases from 12 to 81, and this increase is significant at the 99% level [*Cox*, 1969]. The mean directions of four locations were not used in the above fold test because, as will be discussed in the following sections, these four locations may have experienced tectonic rotations independent of these seven locations. However, by calculating the dispersion of only the inclination values [*McFadden and Reid*, 1982], all eleven location means can be used in a "fold test." The mean of 11 uncorrected inclinations yields a $\kappa = 6$, whereas after structural correction, $\kappa = 53$. This increase is significant at the 99% level. The folding of the Vasquez Formation was initiated immediately after deposition [*Oakeshott*, 1958]. Given the magnetic stability and the positive fold test, the magnetization of the Vasquez volcanics most likely is primary.

Directional Information

The structurally uncorrected and corrected mean paleomagnetic directions for 68 volcanic flows are shown in Figure 6. The results from five units are discarded because their α_{95} values are greater than 20°, and one unit is discarded because of a poorly defined structural correction. The uncorrected directions are highly scattered and yield a very low kappa of 6. The structurally corrected paleomagnetic directions also are scattered. However, with only one exception, the inclinations are all positive and range between 29° and 69.4°. The wide scatter in directions is attributed mainly to the extreme variation in declination which in turn has probably resulted from tectonic rotation about a vertical axis. To better understand the nature of this declination anomaly, the Vasquez Formation was divided into three regions: a central, western, and eastern region.

Central Vasquez (Locations VL, VM, VN, VO, VS, VT, VV)

The structurally corrected directions of 43 flows from seven locations in the central outcrop area of the Vasquez Formation are shown in Figure 7. The clustering of directions is much tighter than for the entire data set (Figure 6). These directions suggest a regional rotation of about 35° clockwise (to be calculated below). However, there are two anomalous characteristics of this data set: (1) the clustering of the majority of units appears to be bimodal and (2) some of the flows still have directions that are highly divergent from the main clustering of directions.

At location VL, 30 flows spanning over 800 m of the volcanic section were analyzed. In Figures 8*a* and 8*b* the inclinations and declinations of these flows are plotted against stratigraphic height. From the base of the volcanic section, the observed declinations decrease rapidly and systematically upsection from over 100° to −60° (Figure 8*a*). Over this same interval, the inclinations are within 15° of the location mean value, with the exception of the prominent "spike" at a stratigraphic height of about 300 m. Two explanations are suggested for the anomalous directions at the base of the volcanic section: (1) All 9

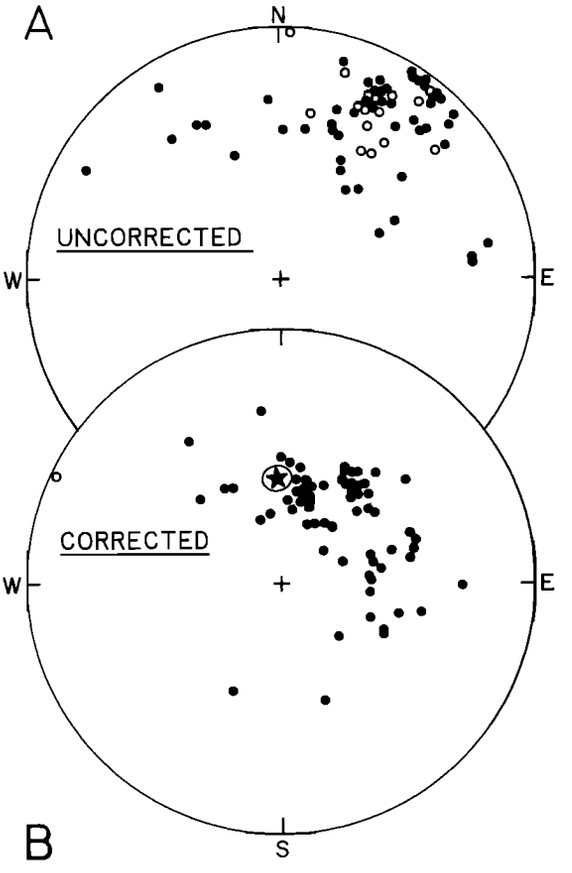


Fig. 6. Equal-area stereoplots of 68 unit mean directions of the Vasquez Formation. (*a*) Structurally uncorrected; (*b*) Structurally corrected. Open circles are upper hemisphere; dots are lower hemisphere; star is 15 Ma reference direction with its associated α_{95} [*Irving*, 1979].

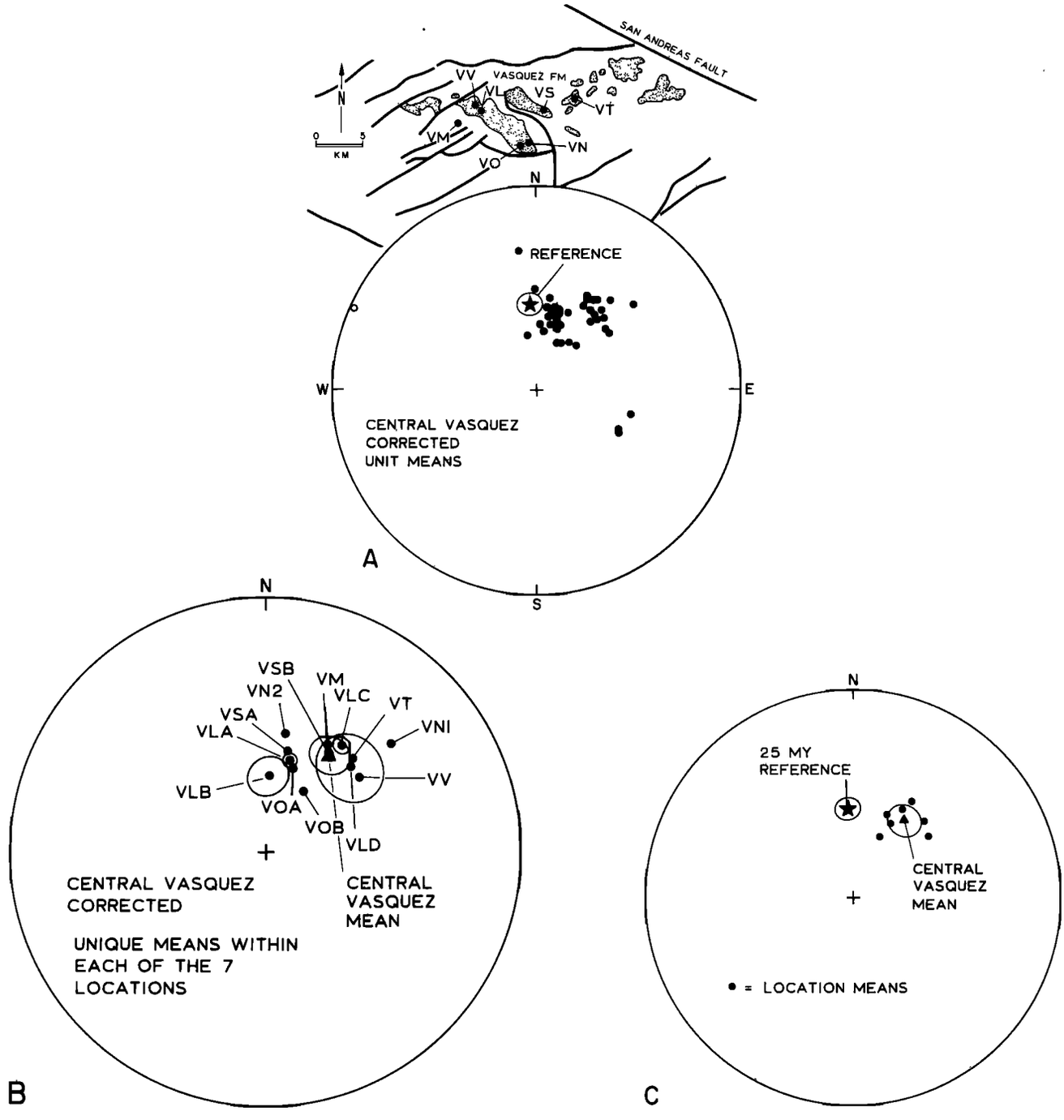


Fig. 7. Equal-area stereoplots of (*a*) structurally corrected unit directions of the central Vasquez region (VL, VM, VN, VO, VS, VT, and VV); (*b*) structurally corrected mean directions of the unique time groups (see text for definition) within each of the seven central Vasquez locations. All mean directions generally conform to the bimodal distribution of directions observed at VL; (*c*) mean of the central Vasquez region (triangle) and its discordance with the 25 Ma reference direction (star). All points are on the lower hemisphere.

basal flows record a geomagnetic anomaly (i.e. reversal or excursion). The "spike" observed in both the inclination and declination values at the 300 m stratigraphic height is most easily attributed to a geomagnetic anomaly. The smooth decrease in declinations from the base of the section into this spike may suggest that the entire declination anomaly is due to one event—a reversal or an excursion. (2) The lack of anomalous inclinations may suggest that the lower six to seven flows record a tectonic rotation about a vertical axis (on the order of 100°) and only the next two or three flows record a geomagnetic anomaly.

The declination difference between the mean direction of the basal three flows and the mean of the upper 21 flows suggests a relative tectonic rotation of 85.5° ± 30.7°. It is doubted that this declination difference represents a regional tectonic rotation primarily because very little time appears to have elapsed between the extrusion of these flows. There are no soil horizons and no sedimentary

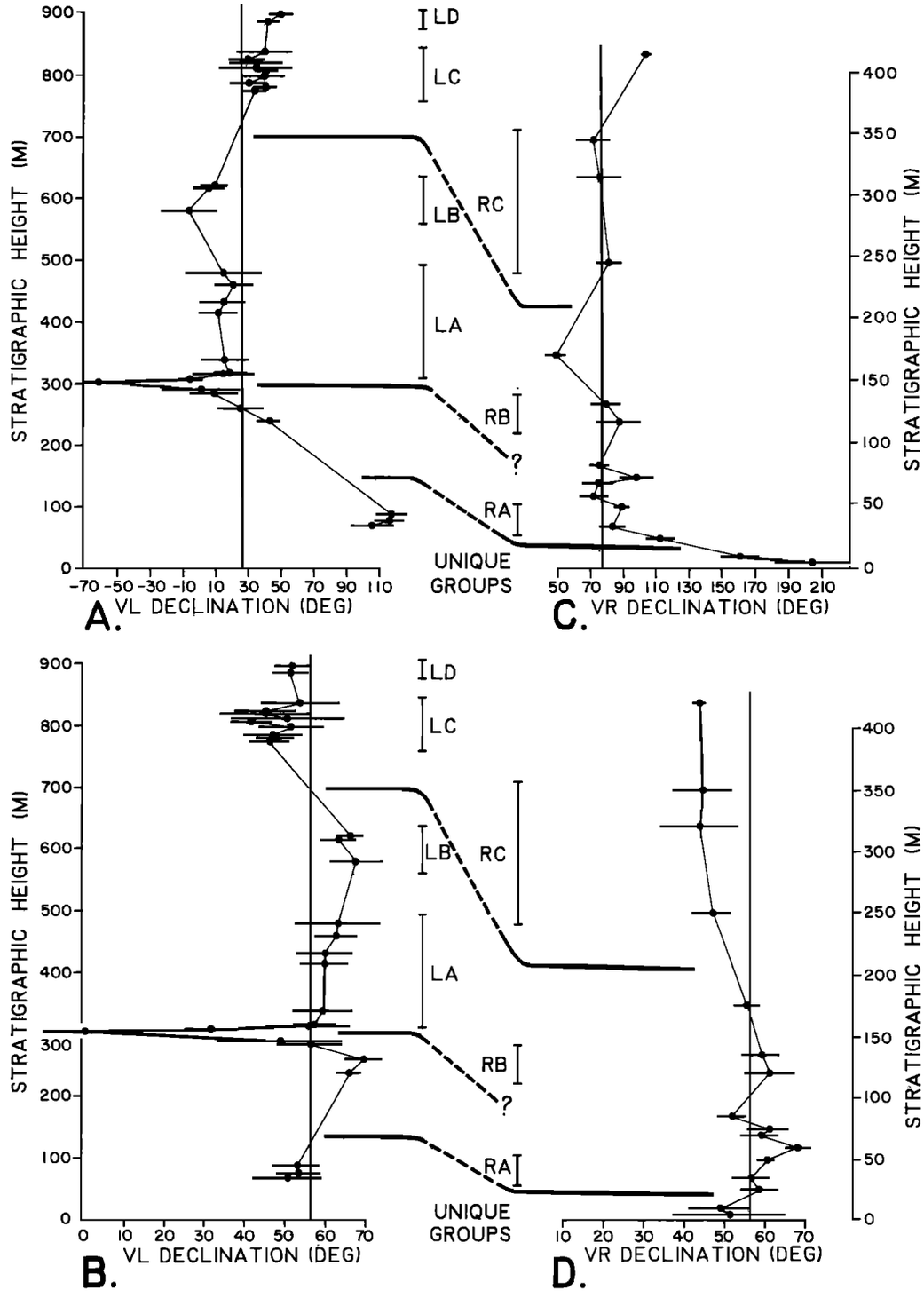


Fig. 8. Paleomagnetic inclinations and declinations in degrees are plotted against stratigraphic height for units at locations VL and VR. Stratigraphic correlations based upon paleomagnetic criteria are shown. Vertical lines are average inclination and declination values. Error bars are the α_{95} as for inclinations and ΔD for declinations. The successions of flows thought to represent rapid outpourings of lava (unique groups) are shown in the central column.

interbeds between the flows with high declinations and those with low declinations. In an active sedimentary basis such as the Soledad basin, which is characterized by the voluminous influx of coarse clastics, it seems unlikely that a significant volcanic hiatus would have been unaccompanied by sediment accumulation. Alternatively, this declination difference may represent shearing within the base of the volcanic pile. Zones of intense fracturing were observed near the base of the volcanic section and may support this hypothesis. Regardless of the correct expla-

TABLE 1. Paleomagnetic Results From the Vasquez Formation

	Number of Samples	Number of Units	Number of Time Units	I_c	D_c	α_{95}	Kappa
			Central: Vicinity 34.48°N 118.23°W				
VL	142	21	4	56.9	27.0	15.2	37.6
VM	6	1	1	49.7	29.6	7.7	76.2
VN	11	2	1	44.7	31.9	78.1	12.5
VO	20	4	2	64.0	24.5	17.3	210.4
VS	29	4	2	54.6	22.7	26.8	88.8
VT	15	1	1	48.3	43.0	2.7	195.7
VV	6	1	1	51.3	51.7	5.7	140.4
Average of seven locations				53.2	33.5	6.7	81.4
Paleomagnetic pole				62.4°N	36.4°W		
			Western: Vicinity 34.49°N 118.35°W				
VR	69	14	10	57.0	77.7	7.6	41.2
Paleomagnetic pole				29.0°N	56.1°W		
			Eastern: Vicinity 34.52°N 118.1°W				
VJ	46	6	6?	52.1	335.0	12.8	28.3
Paleomagnetic pole				69.2°N	154.0°E		
VP	10	2	2	37.0	83.2	44.6	33.5
Paleomagnetic pole				17.0°N	41.9°W		
VQ	5	1	1	65.0	132.3	7.5	105.8
Paleomagnetic pole				2.1°N	87.8°W		

nation, the directions of the basal nine flows are considered to be anomalous and are not used in the determination of the mean direction of location VL.

The magnetic directions of the upper 21 flows at VL have a stratigraphically ordered, roughly bimodal distribution. Four stratigraphically and statistically distinct groups (called "unique groups") of flows can be defined. The criteria for the delimination of these four groups are that the magnetic directions for the units in each group must be indistinguishable at the 99% level from the directions of all the other units in the group and that the direction for the stratigraphically lowest member of each group must be unique at the 99% level from the direction of at least one of the units of the stratigraphically lower group. The 4 unique groups are illustrated in Figures 7*b* and 8. The dispersion of virtual geomagnetic poles (VGP's) for each of the groups are low: the kappas are (1) 466.4, (2) 164.9, (3) 259.6, and (4) 288.4. Thus each of these groupings are thought to represent a temporally distinct rapid outpouring of successive lava flows, and each grouping basically represents a single sampling of the geomagnetic field. The mean direction for location VL was determined by averaging together the means of these four groups and is $I = 56.9°$, $D = 27°$ with $\alpha_{95} = 15.2°$ and $\kappa = 37.6°$ (Table 1).

At locations VO and VS the individual units were divided into unique groups using the criteria outlined above. Also plotted on Figure 7*b* are the mean directions found for the units at VM, VN, VT, and VV. The roughly bimodal distribution of unit directions observed at VL can be seen in the directions from the other locations. This observation suggests that there has been little interlocation tectonic rotation between the seven selected locations of the central Vasquez region.

The mean direction of each sampling location was determined by combining the means of each of the unique group directions within that location (Table 1). The mean direction of the central Vasquez region was defined by the combination of the mean directions for each of the seven locations. This mean direction is $I = 53.2°$, $D = 33.5°$ with $\alpha_{95} = 6.7°$ and $\kappa = 81.4$ (Figure 7*c*).

Relative to the 25 Ma reference direction [*Irving*, 1979], the mean declination of the central region is significantly deflected in a clockwise sense $37.1° \pm 12.2°$ and the inclination is insignificantly shallow ($1.3° \pm 7.8°$).

Some degree of caution should be exercised in interpreting these directions and rotations because the secular variation of the earth's magnetic field may not be entirely averaged out. The degree to which secular variation is averaged can be estimated by a dispersion analysis of the VGP's. From the 34 useful flows of the central Vasquez area (the nine flows at the base of VL were not used), there are at least five temporally independent samplings of the geomagnetic field: each of the four groups at location VL, and the flow at VM which lies stratigraphically above the flows at VL. The dispersion of the VGP's of these five samplings is $\kappa = 27.3$, and at the 95% level of confidence, $10.0 < \kappa < 53.1$ [*Cox*, 1969]. Although the observed κ does not differ at the 95% confidence level from the expected $\kappa = 35.5$ [*McFadden*, 1980], the small number of readings of the geomagnetic field suggests that secular variation may be undersampled. This dispersion analysis may not be entirely valid because it assumes that the VGP's are randomly distributed. This may not be the case for the Vasquez data.

Western Vasquez (location VR)

The flows sampled at VR span a stratigraphic thickness of over 400 m. Inclinations and declinations of the units are plotted as a function of stratigraphic height in Figures 8*c* and 8*d*.

Over a stratigraphic thickness of only 30 m, the four

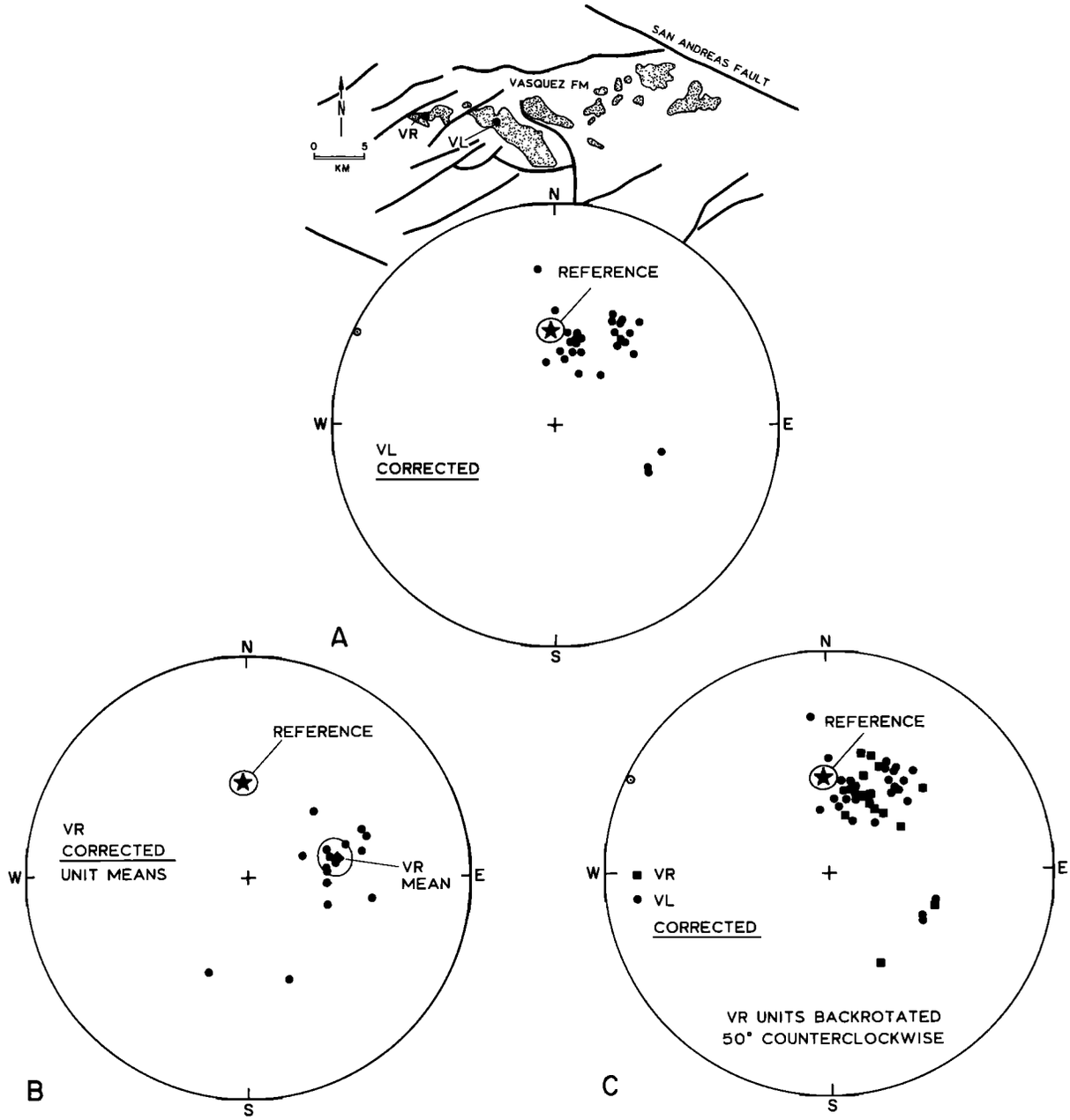


Fig. 9. Equal-area stereoplots of unit directions from (*a*) VL, (*b*) VR, (*c*) VL and VR, but with the directions of VR rotated counterclockwise 50°. In Figure 9*c* the two data sets overlap quite well and there is some correspondence between the units with the highly clockwise-deflected declinations.

basal units at VR show a progressive decrease in paleomagnetic declination from about 200° to 80° (Figure 8*c*). This declination anomaly is believed to correlate with that observed at location VL (Figure 8*a*) and also may be due to either a geomagnetic anomaly or a structural rotation. Some zones of fracturing were observed near the base of the volcanic section at VR. However, the origin of this declination anomaly (as well as that at VL) currently is not known.

In calculating the mean direction of location VR, the directions of the basal two units were rejected because they are far away from the main grouping of directions. For location VR the number of temporally and stratigraphically distinct samplings of the geomagnetic field was determined using the criteria outlined for location VL. Of the 14 flows above the two basal flows, there are 10 unique samplings of the geomagnetic field. Units 4 and 5, 10 and 11, and 13 through 15 have been combined into three unique groups (Figure 8). The mean direction of VR defined from these 10 unique samplings is $I = 57.0°$, $D = 77.7°$, with $\alpha_{95} = 7.6°$ and $\kappa = 41.2$.

The dispersion of the VGP's of the 10 temporally unique directions of location VR give a $\kappa = 25$, and at the 95% level of confidence, $12.5 < \kappa < 41.9$. This κ is not significantly different from the expected $\kappa = 35.5$ [*McFadden*, 1980]. Secular variation appears to be sampled adequately at location VR.

From a comparison of the stratigraphic distribution of

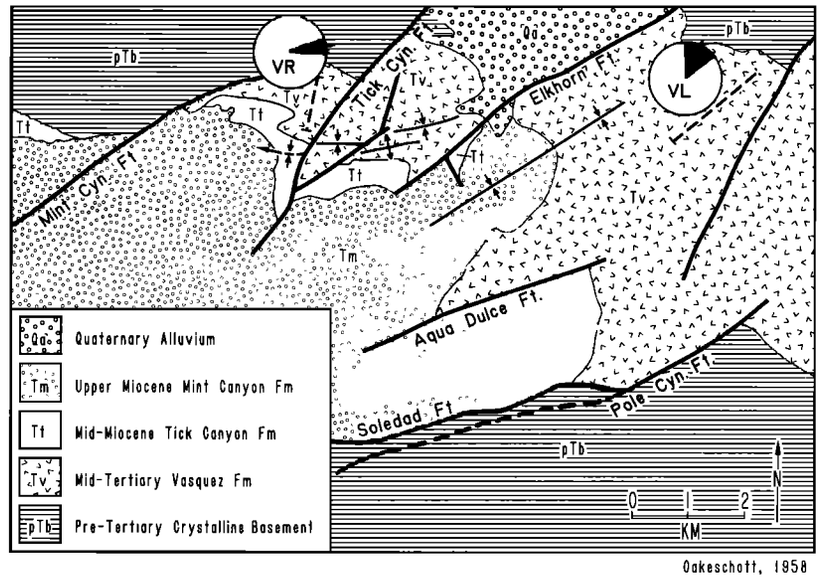


Fig. 10. Geologic map of region around locations VL and VR [*Oakeshott*, 1958]. The mean paleomagnetic directions and their associated uncertainties at locations VR and VL are shown by the sector symbols. The approximately 50° relative rotation of the mean paleomagnetic directions of VL and VR is manifested geologically by the change in fold hinge orientations between VL and VR.

directions from VR in the western Vasquez area (Figures 8*c* and 8*d*) to the stratigraphic distribution of the directions from VL in the central Vasquez area (Figures 9*a* and 9*b*) the following two observations are made. (1) The stratigraphic sequence sampled at VR is largely the same as that sampled at VL. The proposed stratigraphic correlation between the two locations is shown in Figure 8. This correlation is based primarily on the large upsection decrease in paleomagnetic declination at the bases of both VR and VL (Figure 8*a* and 8*c*); the paleomagnetic inclinations in the middle parts of VR and VL generally are higher than the mean values (Figures 8*b* and 8*d*); and the paleomagnetic inclinations of the upper parts of VR and VL generally are lower than the mean values (Figures 8*b* and 8*d*). The prominent "spike" in the inclination and declination values of VL is not seen at VR. This is not surprising in that the volcanic member is only half as thick at VR, and the units which record the spike at VL may not be present at location VR. The above correlation scheme is not as apparent from the distribution of the declinations (above the basal anomalous declinations). This may be attributed to small errors introduced into the declinations by the application of the structural corrections. The paleomagnetic inclination values are not as susceptible to errors induced by the bedding correction because the inclination is fixed relative to the bedding. This correlation of two stratigraphic sections separated by about 10 km reinforces the conclusion that these lavas retain the original TRM. (2) The difference in the mean declinations between locations VL and VR is 50° ± 32°; this suggests that VR has rotated clockwise relative to VL.

Unit mean directions at locations VL and VR are shown in Figures 9*a* and 9*b*. In Figure 9*c* the data sets from VL and VR have been plotted on the same stereonet, but the directions of VR have been backrotated 50° counterclockwise about a vertical axis. With this relative rotation removed, the two data sets overlap extremely well. A 50° clockwise tectonic rotation of VR relative to VL (and the central Vasquez region) is reasonable in that the trends of the major fold axes in the vicinity appear to be locally rotated by roughly 45° relative to one another. As is shown in Figure 10, a major syncline near VL has a

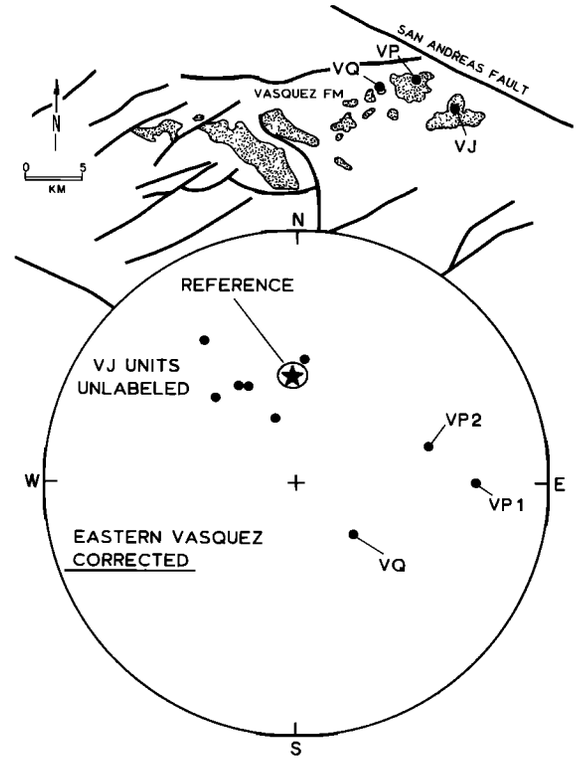


Fig. 11. Equal-area stereoplot of structurally corrected unit mean directions of the eastern Vasquez region (VJ, VP, VQ). Symbols as in Figure 6.

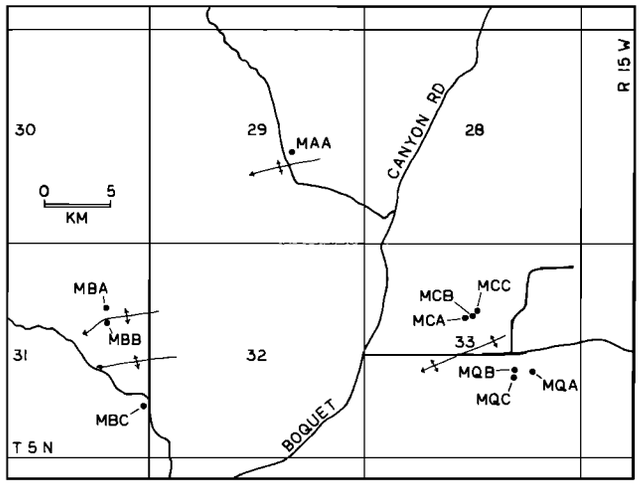


Fig. 12. Paleomagnetic sampling locations of the Mint Canyon Formation. The location of this region is shown in Figure 2.

northeast-southwest trend and plunges to the southwest, but at VR the major syncline has an east-west trend and plunges to the west.

The middle to upper Miocene Mint Canyon and Tick Canyon formations overlie the Vasquez beds with an angular unconformity [*Oakeshott*, 1958], suggesting that the folding and differential rotation are of middle Miocene age.

The extent of the highly clockwise rotated Western Vasquez region is unknown. Directly to the west, the structural trends of the Vasquez Formation are masked by the overlying Mint Canyon Formation. However, the strata of the Western Vasquez location have vertical to slightly overturned dips which may indicate very localized but intense rotation and deformation.

Eastern Vasquez (locations VJ, VP, VQ)

The mean directions of nine flows from the three easternmost locations are shown in Figure 11. The declinations are highly scattered. Six flows from location VJ define a mean direction of $I = 52.1°$, $D = 335.0°$, with $\alpha_{95} = 12.8°$ and $\kappa = 21.1$ (Table 1). The remaining three flows, from locations VP and VQ, have nearly easterly declinations of 74.7°, 90.1°, and 132.3°. These highly divergent declinations may indicate either relative tectonic rotation or a geomagnetic anomaly. Right shear adjacent to the San Andreas fault may have caused clockwise distortion analogous to that reported along the Clemens Well fault of southern California [*Terres et al.*, 1980; *Terres*, 1984]. The counterclockwise-deflected declinations of location VJ may be attributed to shear adjacent to one of the northeast trending left-lateral faults of the San Gabriel block, or the proposed regional counterclockwise rotation of the Mojave Desert region [*Garfunkel*, 1974; J.L. Morton and J.W. Hillhouse, unpublished manuscript, 1983). Alternatively, if these units record anomalous field directions, they may correlate with similarly anomalous directions for observed units in the central (VL) and western (VR) Vasquez regions. Unfortunately, rock exposures in the eastern region are poor and discontinuous, and existing geologic mapping does not allow for discrimination between the above possibilities.

PALEOMAGNETISM OF THE MINT CANYON FORMATION

Paleomagnetic sampling of the Mint Canyon Formation was conducted exclusively on the tuff beds from the upper part of the formation. From four locations, 65 samples from 10 sites were collected (Figure 12). Two types of tuff were collected. At locations MBA, MBB, and MBC, gray, friable, vitric tuffs were sampled. All of the other samples are from well-lithified white tuffs.

White Tuffs

The white tuffs yielded stable magnetic directions when subjected to af demagnetization and were primarily cleaned using this technique. Typical demagnetization behavior is illustrated in Figure 13*a*. Demagnetization in fields up to 30 mT removed a component that is aligned with the present earth's field (probably a VRM). Demagnetization in fields between 30 and 80 mT generally yielded a stable direction. In many samples, demagnetization to 80 mT produced less than a 50% drop in the intensity of magnetization, indicating that the magnetic phase has a high coercivity and is relatively fine-grained (in the pseudo-single domain to single domain range).

Representative samples were subjected to thermal demagnetization and typical behavior is illustrated in Figure 13*b*. At temperatures between 250°C and 500°C, the magnetic directions were stable. The stable directions obtained by thermal demagnetization are virtually identical to those obtained by af - demagnetization (Figures 13*a* and 13*b*). The stable directions are carried by either magnetite or fine-grained titanohematite. The samples only yielded stable magnetizations below the Curie temperature of magnitite (578°C), and there is an extreme drop in the magnetic intensity between 350° and 575°C (Figure 13*c*).

The characteristic magnetization of the white tuffs was acquired prior to the folding of the Mint Canyon Formation, as illustrated by a fold test shown in Figure 14*a*. After correction for folding, the directions of correlative tuffs (see Figure 12) are not significantly different.

The age of magnetization of the white tuff beds is constrained to be late Miocene to early Pliocene. The zircon fission track dates of 10 and 11 Ma from these tuff beds provide an older age limit on their magnetization. The younger limit of the magnetization is the age of the folding of the Mint Canyon Formation in the Soledad basin. This folding occurred prior to late Pliocene time and was probably initiated in early Pliocene time [*Stitt*, 1982].

Directional Analysis

The unit mean directions of the tuffs of the Mint Canyon Formation are listed in Table 2. The corrected directions of the six acceptable white tuff units are shown in Figure 15. Three gray tuff beds are rejected because the magnetization is of post-folding age. Unit MQA was rejected because of a poorly defined structural correction.

Five of the acceptable site means are reversely magnetized, and one is normally magnetized. The one normally magnetized direction (MCB) is from a tuff bed which is stratigraphically between two reversed beds (MCA and MCC).

To define the formation mean direction, the directions of units MQB and MCC, and units MQC and MCA first

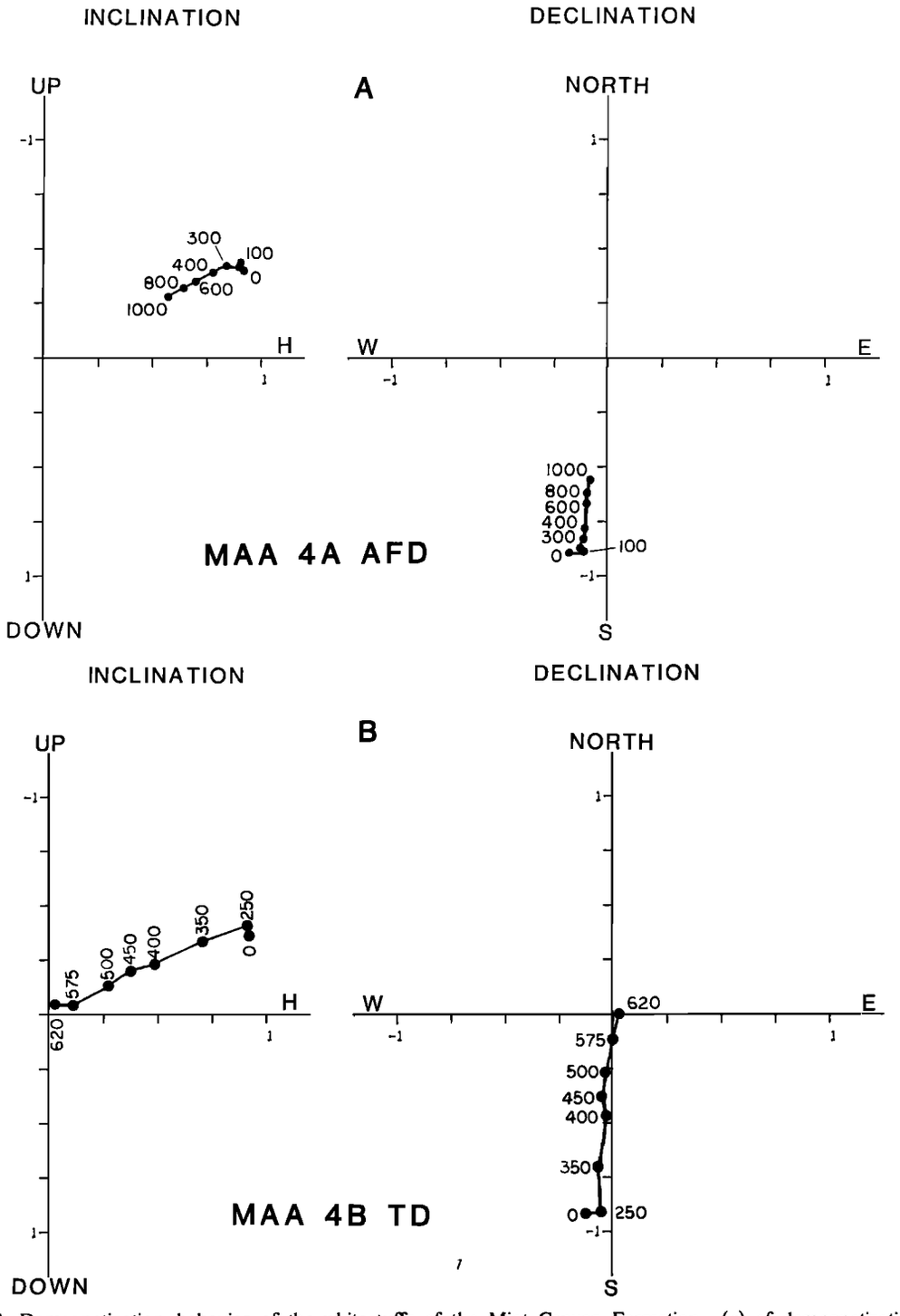


Fig. 13. Demagnetization behavior of the white tuffs of the Mint Canyon Formation. (*a*) af demagnetization of sample MAA 4A. The direction is stable in demagnetization fields greater than 30 mT. (*b*) Thermal demagnetization of MAA 4B. Numbers are demagnetization level in degrees Centrigrade. Direction is stable between 200°C and 500°C. (*c*) Thermal demagnetization of magnetic intensity of representative white tuffs. A large intensity loss between 300°C and 580°C indicates the presence of magnetite.

were combined since they are from the same beds and then the two resultant directions were combined with the directions of the remaining two units (MAA and MCB). The direction obtained is $I = -39.8$, $D = 164.3$ with $\alpha_{95} = 22.9$ and $\kappa = 17.0$ (Table 2). To test for the extent to which secular variation is averaged, the dispersion of the VGP's of the four units was calculated. The resultant pole has a $\kappa = 17.9$, and at the 95% level of confidence ($5.8 < \kappa < 36.6$), κ is not significantly different from the predicted $\kappa = 35.5$ [*McFadden*, 1980]. This suggests that the dispersion of directions can be ascribed to secular variation. However, because of the small number of meas-

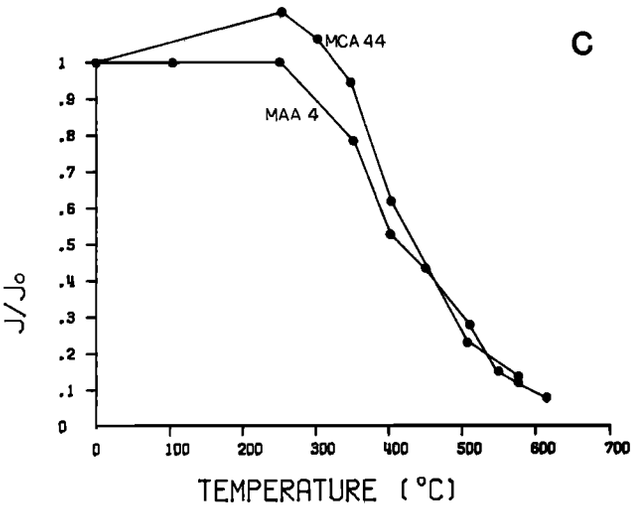


Fig. 13. (continued)

ured units, it is doubted that secular variation is adequately averaged, and again caution should be observed in the use of the mean paleomagnetic direction.

Relative to a magnetic pole aligned with the present spin axis of the Earth, the apparent tectonic rotation of the tuff beds is $-16° \pm 30°$ (in a counterclockwise sense) and the inclination is shallow by $14° \pm 23°$. The present spin axis is used as a reference direction because the position of North America with respect to the spin axis has not changed significantly over the last 10 Ma [*Irving*, 1979].

Tectonic Rotations of the San Gabriel Block

Paleomagnetic data from the San Gabriel block and environs support a two-stage model of regional rotations; in early to middle Miocene time, approximately 50° of clockwise rotation, followed in Pliocene to Recent time by approximately 15° of counterclockwise rotation.

Evidence for the younger counterclockwise rotation comes mainly from the Mojave province, located directly east of the San Gabriel block. There, a paleomagnetic study of upper Miocene basalt flows yielded a discordant pole which suggests $15° \pm 11°$ of counterclockwise rotation [*Burke et al.*, 1982; J.L. Morton and J.W. Hillhouse, unpublished manuscript, 1983]. Counterclockwise deflection of the mean declination of the Mint Canyon Formation ($-16° \pm 30°$) is consistent with the rotation suggested from the Mojave rocks. Furthermore, the San Andreas fault, a post-Miocene structure [*Crowell*, 1981], has a prominent counterclockwise bend where it transects the Mojave and San Gabriel regions; this observation also supports the idea of post-Miocene counterclockwise rotation in the region.

The declinations and associated uncertainties observed in the Vasquez Formation are summarized in Figure 16. The mean declination of the central Vasquez region is discordant in a clockwise sense by $37° \pm 12°$ and is believed to record a net regional rotation of this amount because the seven locations of this region cover an area of greater than 50 km^2, they are found in simple but different local structural settings, and they have relatively uniform paleomagnetic directions. This clockwise rotation may have occurred in early to middle Miocene time since Vasquez beds are rotated clockwise whereas upper Miocene Mint Canyon beds are not. Assuming that the San Gabriel block did undergo approximately 16° of post-Miocene counterclockwise rotation, the actual Miocene regional clockwise rotation would have been about 53° (16° + 37°).

The paleomagnetic sampling locations in the western and eastern Vasquez regions are structurally complex (dips to 90°) and anomalous declinations observed are not consistent either from location to location or with those of the

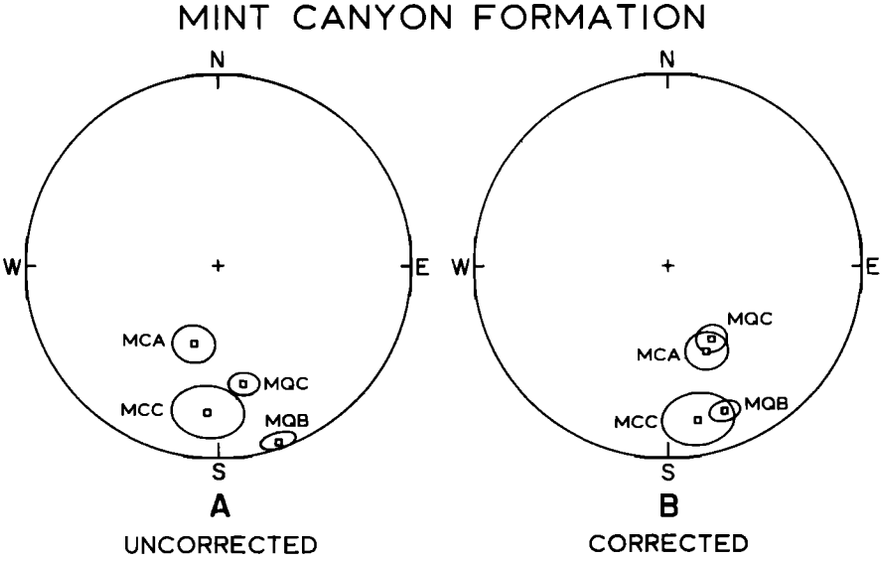


Fig. 14. Equal-area stereoplots illustrating a positive fold test for the white tuffs of the Mint Canyon Formation. All symbols are upper hemisphere mean directions with associated α_{95}. (*a*) Uncorrected. (*b*) Structurally corrected. MCA is from the same bed as MQC, and MCC is from the same bed as MQB. The coincidence of the corresponding units is significantly improved after correction.

TABLE 2. Paleomagnetic Results from the Mint Canyon Formation

	Number of Samples	Number of Units	Number of Time Units	I_c	D_c	α_{95}	Kappa
MQB+MCC	12	1	1	−17.2	165.4	37.9	45.6
MQC+MCA	9	1	1	−49.6	155.9	37.5	46.4
MAA	6	1	1	−34.5	180.1	6.9	94.4
MCB	5	1	1	62.0	316.9	12.2	40.4
Average of four units				−39.8 (39.8	164.3 344.3)	22.9	17.0
Paleomagnetic pole				71.8°N	114.9°E		

Location, vicinity 34.48°N 118.45°W

central region and are thought to record either local shear-related rotation or anomalous magnetic field behavior.

Tectonic Model

A schematic model for the post-early Tertiary tectonic development of the San Gabriel block is illustrated in Figure 17 and is described below.

In early Tertiary time the crust was broadly stratified into an upper plate of pre-Tertiary crystalline basement and a lower plate of Pelona Schist [*Ehlig*, 1981] (Figure 17*a*). These two plates are separated by the presumed Paleocene Vincent thrust [*Ehlig*, 1981].

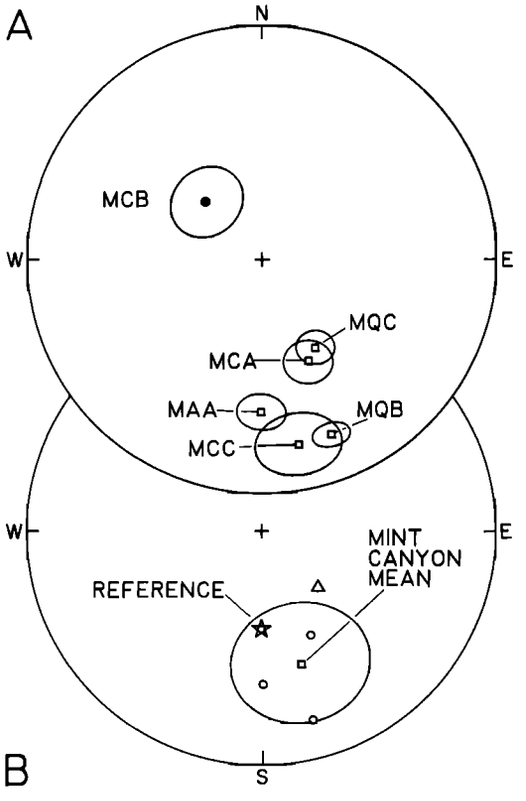


Fig. 15. (*a*) Equal-area stereoplot of structurally corrected unit directions of white tuffs from the Mint Canyon Formation. (*b*) Mean direction of the Mint Canyon Formation white tuffs. MCB is projected through the origin (triangle). MQC and MCA are combined; MCC and MQB are combined. Open symbols are upper hemisphere, solid symbols are lower hemisphere.

In late Oligocene to early Miocene time, crustal extension resulted in broad warping of the crystalline basement and in the development of the Vasquez basins (Figure 17*b*). The Vasquez Formation was deposited in elongate normal-fault controlled basins which presently are oriented east-northest [*Bohannon*, 1975]. Restoration of the net 35° clockwise rotation inferred for the Vasquez volcanics gives these basins an originally north-northeast orientation. The inferred crustal extension was in a generally east-west direction. The association of basaltic volcanism, continental red bed and evaporate deposition, and grabens resulting from an east-west extension suggests that the Vasquez basins record the initiation of basin and range tectonics in southern California [*Luyendyk et al.*, this issue]. This accords well with the late Oligocene inception of basin and range tectonism observed elsewhere in the southwestern United States [*Christiansen and Lipman*, 1972; *Armstrong and Higgins*, 1973]. In early to middle Miocene time, continental deposition continued with the Tick Canyon and Mint Canyon formations, but these strata overlapped the Vasquez basins and their bounding normal faults (Figure 17*c*).

The approximately 50° clockwise tectonic rotation occurred mainly in middle Miocene time in response to regional dextral shear [*Luyendyk et al.*, 1980] (Figure 17*d*). Plate motion studies suggest that during early and middle Miocene time the San Gabriel block was within the northwest trending dextral transform boundary between the Pacific and American plates [*Atwater*, 1970; *Atwater and Molnar*, 1983]. Dextral shear was accommodated by slip on antithetic left-lateral faults which developed parallel to the north-northeast trending Vasquez basins and by clockwise rotation of the intervening crustal blocks [*Dibblee*, 1977; *Luyendyk et al.*, 1980]. Left-lateral faults are found throughout the San Gabriel block and include the San Antonio, Transmission Line, Magic Mountain, Pole Canyon, Agua Dulce, Elkhorn, Tick Canyon, Mint Canyon, and San Francisquito faults [*Oakeshott*, 1958; *Dibblee*, 1977; *Jennings and Strand*, 1969]. In addition, there are many more similar mapped and unmapped faults [*Sams*, 1964; Remenyi, 1966; *Carter*, 1982]. The inferred time of activity along these faults roughly is synchronous with the inferred period of clockwise rotation. These faults cut the early Miocene Vasquez and Tick Canyon formations but are overlapped by late Miocene beds of the upper part of the Mint Canyon Formation [*Muehlberger*, 1958; *Jahns*, 1940; *Oakeshott*, 1958]. Some renewed left-lateral faulting occurred in Pliocene and Pleistocene time,

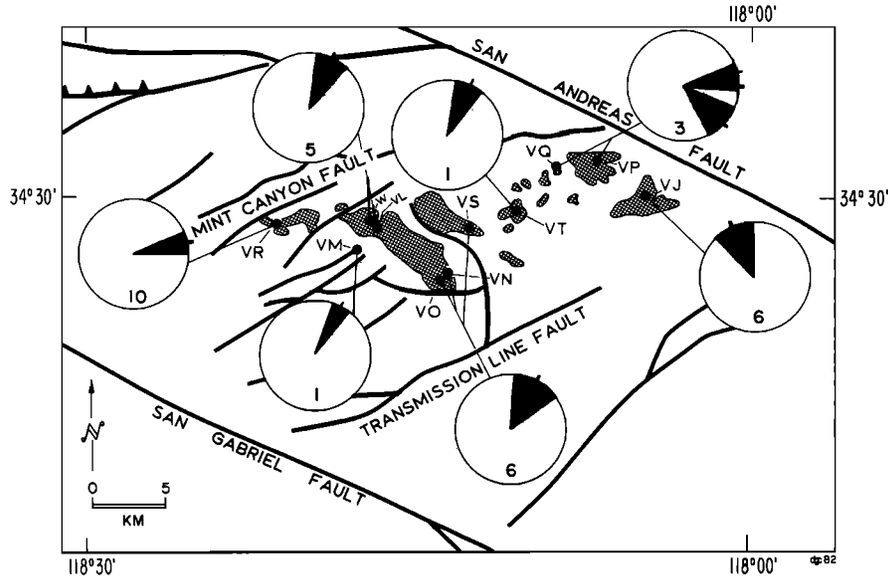


Fig. 16. Map which summarizes paleomagnetic declinations observed in the Vasquez Formation. Mean declinations and associated uncertainties are shown by the darkened sectors. The numbers within the circles are the number of units or unique groups used for each mean direction. The central Vasquez region is characterized by uniform declinations, whereas the eastern and western Vasquez regions have highly variable declinations.

but it was minor [*Oakeshott*, 1958; *Crowell*, 1968]. The maximum left separation observed along any one of these faults is about 6 km [*Oakeshott*, 1958; *Carter*, 1982]. Using geometric models relating dextral shear, fault slip, and block rotation [*Freund*, 1974; *Luyendyk et al.*, 1980], Terres [1984] suggests that the magnitude of the observed left-lateral displacements within the San Gabriel block is compatible with a 50° clockwise rotation.

In order to drive the middle Miocene clockwise rotation the crust and possibly mantle are mechanically stratified into a relatively brittle upper layer which deforms by faulting (and rotation) and is separated by a subhorizontal decollement from a possibly more ductile lower zone which accommodates the dextral shear either by ductile shear, right-lateral strike-slip faulting, or a combination of the two.

Candidates for the decollement in the San Gabriel block include the Vincent thrust and horizontal seismic discontinuities seen in the Transverse Ranges. The seismic discontinuities are observed at depths between 10 and 20 km [*Hadlev and Kanamori*, 1988; *Malin et al.*, 1981].

The Vincent thrust is exposed to the north of the Soledad basin along the south flank of Sierra Pelona and is exposed to the south of the Soledad basin in the eastern San Gabriel Mountains. Interpolation between these exposures suggests that the Vincent thrust lies beneath the Soledad basin and much of the San Gabriel block [*Ehlig*, 1981]. If the early Tertiary Vincent thrust was reactivated in middle Miocene time, it may have provided a suitable decollement above which crustal blocks experienced brittle failure and rotation. The spacing between NE trending left-slip faults generally decreases to the north in the San Gabriel block (Figure 2). Assuming that the relative spacing between faults indicates the relative thickness of the brittle layer (depth to the decollement [*Merzer and Freund*, 1976]), the decollement appears to be ramping up to the north toward the Vincent thrust exposure (Figure 2).

Alternatively, if the Vincent thrust was cut by the mid-Tertiary normal faulting which created the Vasquez basins, the Vincent thrust may have been given quite high structural relief and thereby rendered an ineffective regional decollement. In this case, one of the deeper, seismically observed crustal discontinuities may be the decollement for the rotation.

The clockwise rotation and left-lateral faulting ceased as dextral shear became localized along the northwest trending right-lateral San Gabriel fault in late Miocene time (Figure 17*e*). Activity on the San Gabriel fault probably began about 12 m.y. ago [*Crowell*, 1982] at roughly the same time that displacement ended on the northeast trending left-lateral faults. The San Gabriel fault was active until about 5 m.y. ago when it was crosscut by the San Andreas fault [*Crowell*, 1982]. The San Andreas fault has displaced the San Gabriel block about 240 km right laterally from an original position just west of the Orocopia and Chocolate Mountains of southeastern California [*Crowell*, 1981].

In Pliocene to Recent time the San Gabriel block underwent an approximately 15° counterclockwise rotation due to movement into and along the "big bend" of the San Andreas fault [after *Garfunkel*, 1974]. Presumably, the San Andreas fault started out nearly straight, but differential crustal extension in the Basin and Range Province and the southern California continental borderland caused a counterclockwise rotation of the Mojave region and a bending of the adjacent San Andreas fault [*Garfunkel*, 1974]. According to *Garfunkel* [1974] the counterclockwise rotation might be as much as 30°, but this magnitude of rotation as yet has not been demonstrated by paleomagnetic studies.

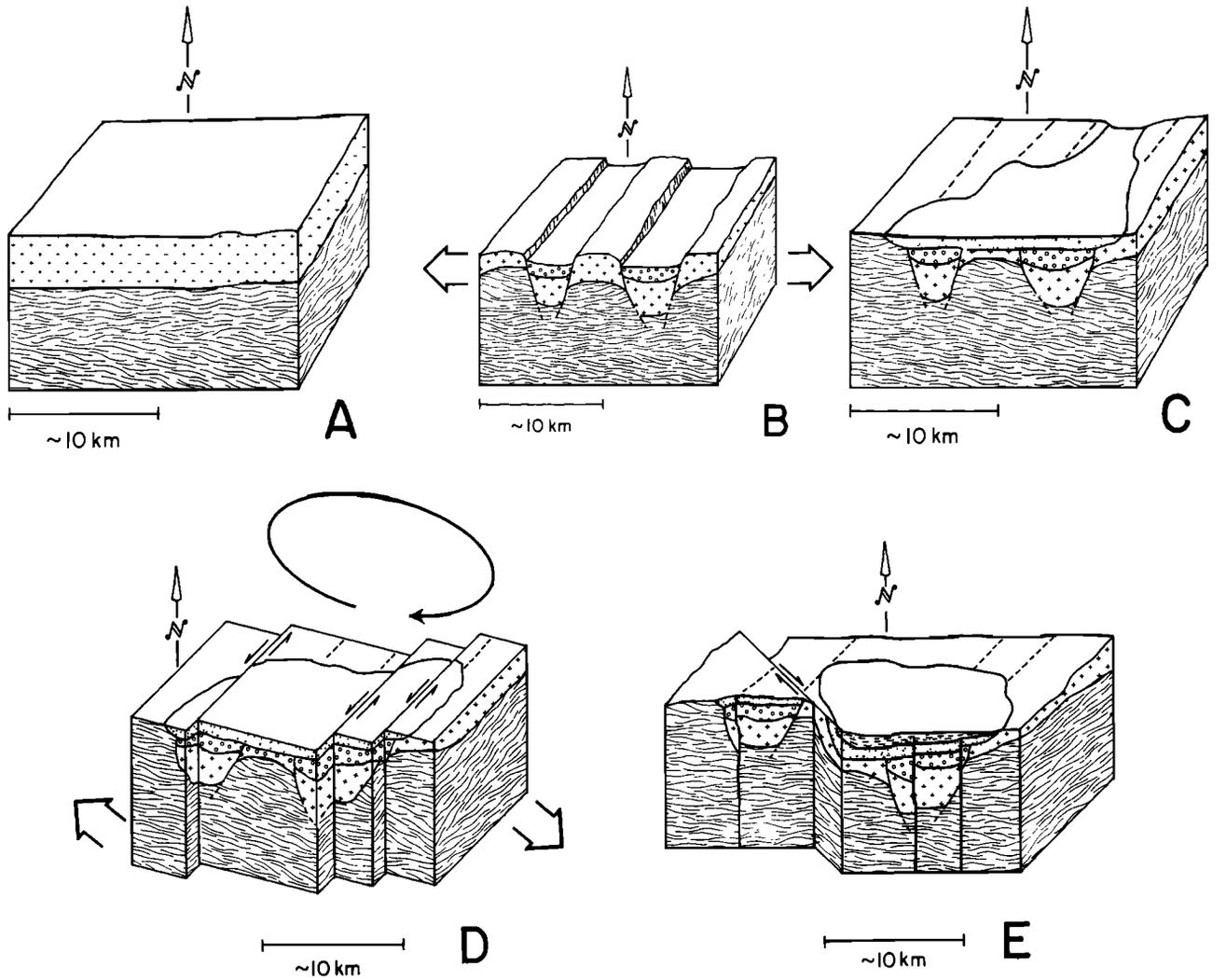


Fig. 17. Schematic model of development of the uniform approximately 35° net clockwise rotation of the San Gabriel block. (*a*) Early Tertiary. Pre-Tertiary crystalline basement rocks structurally overlie schist (Pelona Schist). (*b*) Mid-Tertiary crustal extension. Development of mid-Tertiary continental basin deposits (Vasquez Formation). (*c*) Early Miocene unconformable overlap of mid-Tertiary basins (Tick Canyon Formation). (*d*) Middle Miocene clockwise rotation and development of NE trending left-lateral faults in response to dextral shear. (*e*) Late Miocene end to clockwise rotation and left-lateral faulting. Initiation of the right-lateral San Gabriel fault.

DISCUSSION

The tectonic model presented is considered speculative and requires more complete documentation. The paleomagnetic studies of the Vasquez and Mint Canyon formations do not average completely secular variation and do not sample a sufficiently large area to unambiguously define regional rotations. These data allow for many possible interpretations, for example, all discordant declinations are attributed to local shear rotations from both right-slip faulting (clockwise rotation) and left-slip faulting (counterclockwise rotation); alternatively, an approximately 90° clockwise regional rotation may be indicated by the western Vasquez result (VR) and locations VQ and VP in the eastern Vasquez area, with lesser clockwise deflections indicative of superimposed regional and local counterclockwise rotations.

The interpretation that all rotations are due to local shearing is not favored because the central Vasquez data set has discordant declinations which are somewhat uniform over a large area and at seven separate locations. Likewise, the Mint Canyon result used data from three locations spread over a distance of about 2.5 km. Although Vasquez locations VR, VQ, and VP all have highly clockwise deflected declinations, these locations are very complex structurally (dips of 70° to 90° with poorly defined fold plunges) and are most suggestive of local intense deformation rather than regionally acting rotation. The proposed 50° middle Miocene regional rotation is favored because this rotation correlates well with the middle Miocene age and magnitude of left-lateral faulting in the San Gabriel block (see previous discussion). Finally, a 50° clockwise rotation is consistent with the middle Miocene rotation of the Western Transverse Ranges as inferred from the paleomagnetism of the Monterey Formation [*Hornafius*, this issue].

To further test the proposed regional middle Miocene

rotation, paleomagnetic studies of the Oligocene to early Miocene Vasquez red beds in the Charlie Canyon and Texas Canyon basins and early Miocene intrustives in the southern San Gabriel block could be conducted. Additional paleomagnetic study of the Mint Canyon Formation and younger units (Castaic Formation, Ridge Basin Group) would test the proposed young counterclockwise rotation.

The *Luyendyk et al.* [1980] model originally predicted middle Miocene clockwise rotations of up to 120°. This San Gabriel study along with the Western Transverse Ranges paleomagnetic studies [*Hornafius*, 1984, this issue; *Kamerling and Luyendyk*, this issue] suggest the middle Miocene rotation was only of the order of 50° and that in late Miocene to Recent times this regional rotation has been overprinted more locally by both clockwise and counterclockwise rotations.

CONCLUSIONS

An approximately 50° clockwise middle Miocene tectonic rotation has been observed in both the western Transverse Ranges [*Kamerling and Luyendyk*, 1979; *Kamerling and Luyendyk*, this issue; J.S. Hornafius et al., unpublished manuscript, 1985] and the San Gabriel block. Thus, the extent of the terrane which experienced major Miocene clockwise tectonic rotations can be traced as far east as the San Andreas fault. As predicted by *Luyendyk et al.* [1980], this Miocene clockwise-rotated terrane also may extend further eastward into the region of southeastern California from which the San Gabriel block was offset by the post-Miocene San Andreas fault.

The general uniformity in the magnitude of the middle Miocene rotations in the western Transverse Ranges and the San Gabriel block argues for uniformity in the tectonic mechanism by which the rotations occurred. The tectonic model developed here for the San Gabriel block is similar to that developed by *Luyendyk et al.* [1980] for the entire Transverse Ranges. In this model, regionally distributed dextral shear between the Pacific and American plates was accommodated by the development and subsequent clockwise rotation of originally north-northeast- to north trending left-lateral faults. The formation of these left-lateral faults presumably was controlled by the prior existence of north-northeast to north trending normal faults which resulted from a mid-Tertiary crustal extension [*Terres*, 1982; *Luyendyk et al.*, this issue]. The geology of the San Gabriel block together with the paleomagnetism of the Vasquez and Mint Canyon formations support this tectonic model and further suggest that this model is plausible for the Miocene clockwise rotation of much of the Transverse Ranges.

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